



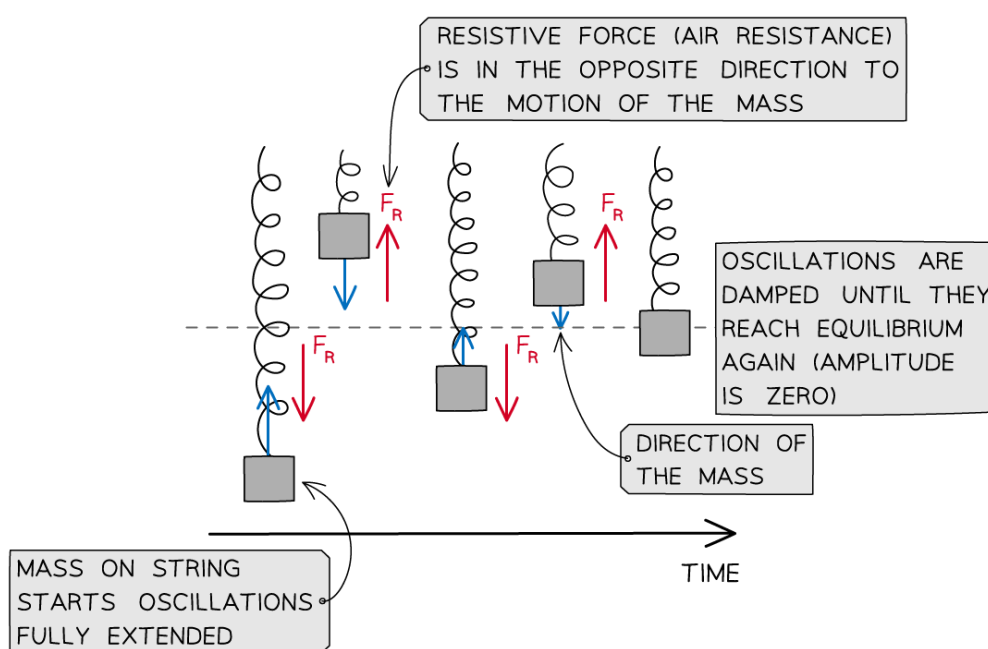
Types of Damping

- In practice, all oscillators eventually stop oscillating
 - Their **amplitudes** may decrease rapidly or gradually
- This happens due to **resistive forces**, such as friction or air resistance, which act in the **opposite** direction to the motion of an oscillator
- Resistive forces acting on an oscillating simple harmonic system cause **damping**
 - These are known as **damped** oscillations

- Damping is defined as:

The reduction in energy and amplitude of oscillations due to resistive forces on the oscillating system

- Damping continues until the oscillator comes to rest at the equilibrium position
- A key feature of simple harmonic motion is that the **frequency** of damped oscillations **does not change** as the amplitude decreases
 - For example, a child on a swing can oscillate back and forth once every second, but the time remains the same regardless of the amplitude



Damping on a mass on a spring is caused by a resistive force (F_R) acting in the opposite direction to the motion (blue arrow). This continues until the amplitude of the oscillations reaches zero

Types of Damping



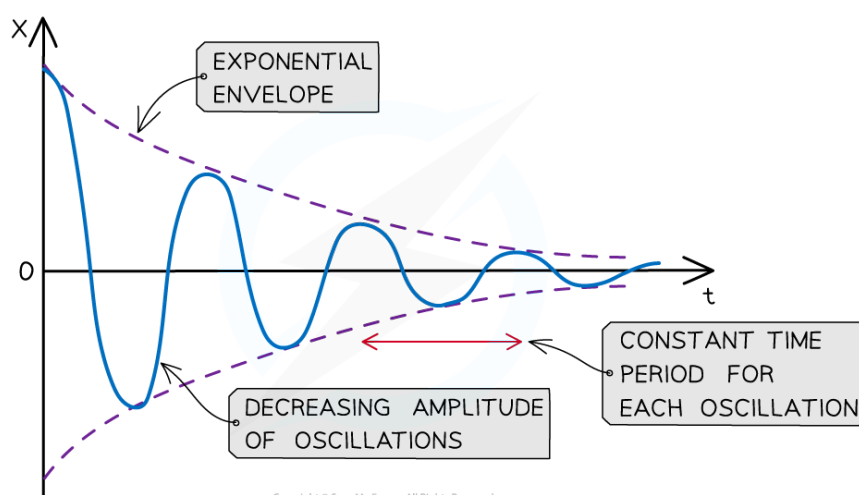
Your notes

- There are three degrees of damping depending on how quickly the amplitude of the oscillations decreases:

- Light damping
- Critical damping
- Heavy damping

Light damping

- When oscillations are lightly damped, the amplitude **does not decrease linearly**
 - It **decays exponentially** with time
- When a lightly damped oscillator is displaced from the equilibrium, it will oscillate with **gradually decreasing** amplitude
 - For example, a swinging pendulum decreases in amplitude until it comes to a stop



A graph for a lightly damped system consists of oscillations decreasing exponentially

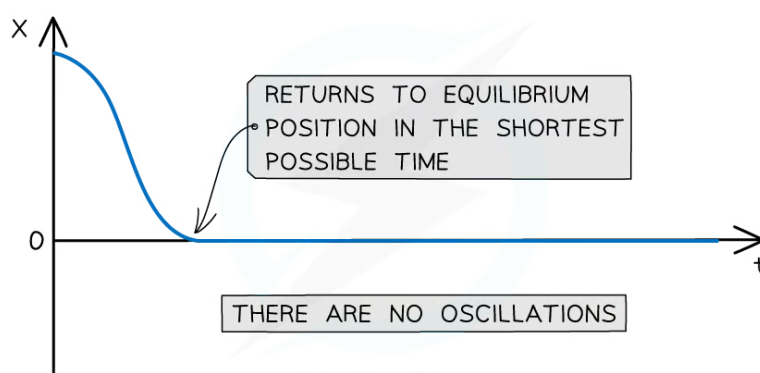
- Key features of a displacement–time graph for a lightly damped system:**
 - There are many oscillations represented by a sine or cosine curve with gradually decreasing amplitude over time
 - This is shown by the height of the curve decreasing in both the positive and negative displacement values
 - The amplitude decreases **exponentially**
 - The frequency of the oscillations remains constant, which means the time period of oscillations must stay the same, and each peak and trough are equally spaced

Critical damping



Your notes

- When a critically damped oscillator is displaced from the equilibrium, it will **return** to rest at its equilibrium position in the shortest possible time **without** oscillating
 - For example, car suspension systems prevent the car from oscillating after travelling over a bump in the road

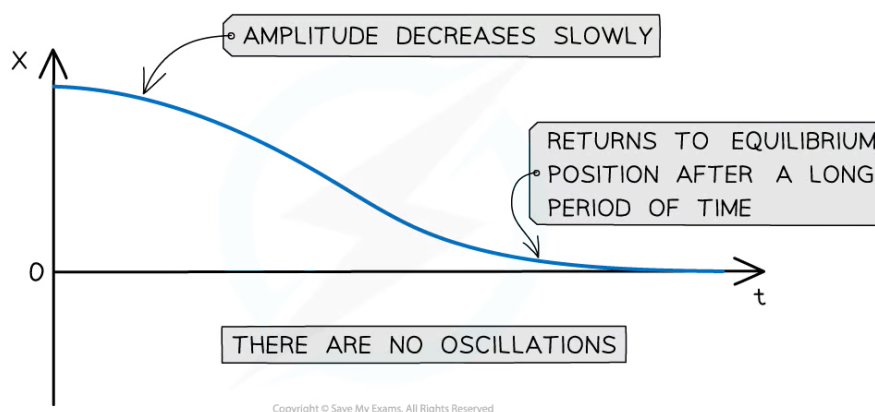


The graph for a critically damped system shows no oscillations and the displacement returns to zero in the quickest possible time

- Key features of a displacement–time graph for a critically damped system:**
 - This system does **not** oscillate, meaning the displacement falls to 0 straight away
 - The graph has a fast decreasing gradient when the oscillator is first displaced until it reaches the x-axis
 - When the oscillator reaches the equilibrium position ($x = 0$), the graph is a horizontal line at $x = 0$ for the remaining time

Heavy damping

- When a heavily damped oscillator is displaced from its equilibrium position, it will take a **long time** to return to its equilibrium position **without** oscillating
- The system returns to equilibrium more slowly than the critical damping case
 - For example, door dampers to prevent them from slamming shut



A heavy damping curve has no oscillations, and the displacement returns to zero after a long period of time



Your notes

■ **Key features of a displacement–time graph for a heavily damped system:**

- There are no oscillations. This means the displacement does not pass 0
- The graph has a **slow decreasing gradient** from when the oscillator is first displaced until it reaches the x-axis
- The oscillator reaches the equilibrium position ($x = 0$) after a **long** period of time, after which the graph remains a **horizontal** line for the remaining time



Worked Example

A mechanical weighing scale consists of a needle which moves to a position on a numerical scale depending on the weight applied.

Sometimes the needle moves to the equilibrium position after oscillating slightly, making it difficult to read the number on the scale to which it is pointing.

Suggest, with a reason, whether light, critical or heavy damping should be applied to the mechanical weighing scale to read the scale more easily.

Answer:

- Ideally, the needle should not oscillate before settling - this means the scale should have either **critical** or **heavy damping**
- Since the scale is read straight away after a weight is applied, ideally, the needle should settle as quickly as possible
- Heavy damping would mean the needle will take some time to settle on the scale
- Therefore, **critical damping** should be applied to the weighing scale so the **needle can settle as quickly as possible** to read from the scale



Examiner Tips and Tricks

Make sure not to confuse **resistive** force and **restoring** force:

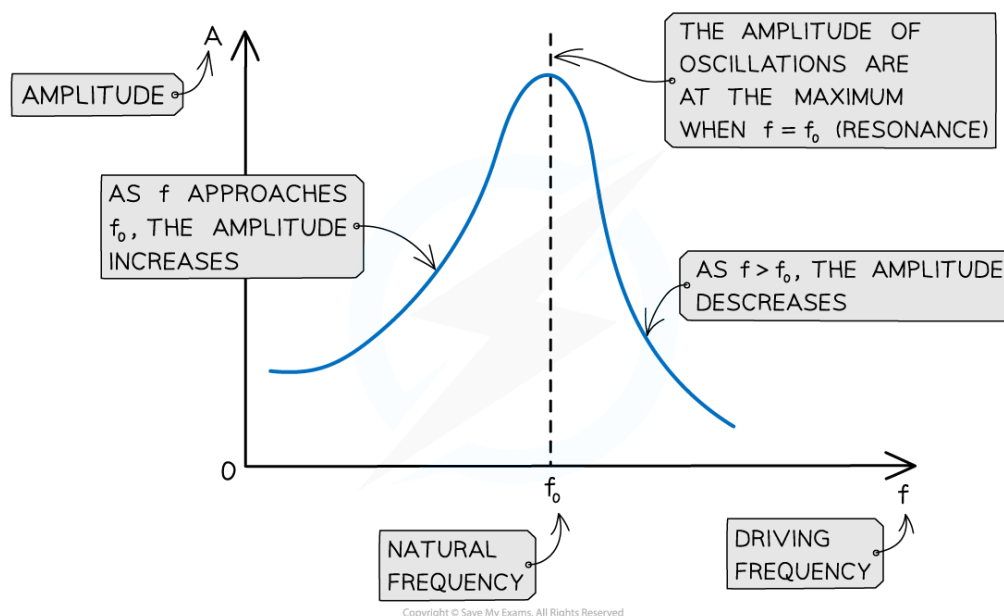
- Resistive force is what **opposes the motion** of the oscillator and causes damping
- Restoring force is what brings the oscillator **back to the equilibrium position**

Effects of Damping

- The effects of damping can be seen on a **resonance curve**
 - This is a graph of driving frequency f against amplitude A of oscillations
 - The graph peaks at the natural frequency f_0 of the oscillator



- It has the following key features:
 - when $f < f_0$, the amplitude of oscillations **increases**
 - at the peak, where $f = f_0$, the amplitude is at its **maximum** - this is **resonance**
 - when $f > f_0$, the amplitude of oscillations starts to **decrease**

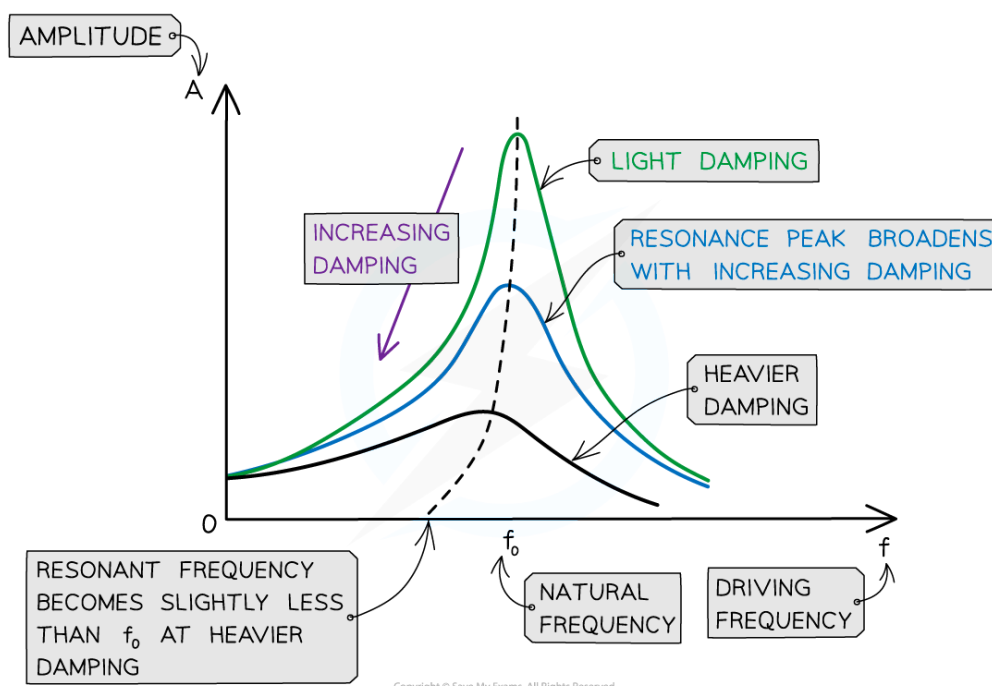


The maximum amplitude of the oscillations occurs when the driving frequency is equal to the natural frequency of the oscillator

- Damping **reduces** the amplitude of resonance vibrations
- The height and shape of the resonance curve will therefore change slightly depending on the degree of damping
 - Note:** the natural frequency f_0 of the oscillator will remain the same
- As the degree of damping is increased, the resonance graph is altered in the following ways:
 - The amplitude of resonance vibrations **decreases**, meaning the peak of the curve lowers
 - The resonance peak **broadens**
 - The resonance peak moves slightly to the **left** of the natural frequency when heavily damped
- Therefore, damping **reduces** the sharpness of the resonance and the amplitude of oscillations at the resonant frequency



Your notes



As damping is increased, resonance peak lowers, the curve broadens and moves slightly to the left



Examiner Tips and Tricks

An exam question can include any damping scenario. Therefore, make sure you can interpret the amplitude-frequency graph depending on the type of oscillation.

Most importantly, the resonant frequency, the frequency of vibrations of the oscillator, can **change** with damping. The natural frequency of an oscillator is the frequency it naturally or normally oscillates. This is the frequency it oscillates at when it is not damped. Therefore, damping **does not affect** the natural frequency.